ANALYTICAL REPORT

Supply Chain Analytics

Service Level and Inventory Intelligence

What the data says

Service quality across the four-warehouse network has eroded steadily over two years while inventory sits unevenly against demand. This report quantifies where service is failing, which suppliers and locations carry the exposure, how concentrated the losses are, and the size of the value pool that disciplined intervention can recover.

79%
of lost-sales value tied to two suppliers

7.6 pts
network fill-rate decline over 24 months

€1.6M
estimated 12-month value pool

82%
of lost sales in the worst 10% of SKU-pairs

Multi-warehouse distribution network | 24 months observed (Jan 2024 - Dec 2025)
Data through 31 December 2025

Synthetic dataset. Financial values are proxy estimates.

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SECTION 1

Executive summary

The operating issue is now decision-grade, not exploratory. Across 120 products, four warehouses, and 12 suppliers over 24 months, the network lost €20.3M of demand to stockouts, worth €8.8M in gross margin at category-level margin assumptions. The monthly fill rate fell from 99.6% in January 2024 to 92.0% in December 2025, a decline of 7.6 percentage points. Over the same window the stockout rate rose from 0.4% to 7.9%. The pattern is a structural drift in replenishment performance, not a set of isolated shocks: the final quarter lost sales ran 3.1x the first-quarter baseline and the final month is the worst in the series.

The loss is concentrated enough to manage surgically. Two suppliers, Supplier 2 and Supplier 3, account for €16.0M of lost-sales value, 79% of the total, and are the only two suppliers in the High risk tier. Their combined on-time rate is 72% versus 90% for the rest of the base, and their lead-time variability is 3.9 days versus 1.3. At SKU-location grain the concentration is sharper still: the worst 10% of the 480 scored pairs carry 82% of all lost sales, and the worst 5% carry 66%.

One product is the cleanest root-cause candidate. Of the 480 product-warehouse pairs scored, only 4 reach the High tier, and all 4 are Health Product 16, one in each warehouse. Combined they represent €530k of recoverable annual value, 62% of the ranked SKU-location pool. The same product fails wherever it is stocked, which makes the working hypothesis a product-level supply, allocation, or planning fault rather than a local warehouse execution issue.

The inventory problem is allocation quality, not aggregate stock. Madrid and Lyon generate 62% of warehouse lost-sales value while Lisbon and Porto hold 59% of inventory value. The fill-rate spread across warehouses is 3.7 points and tracks placement, not total volume. At the same time 22% of SKUs hold 30 or more days of supply against a 17-day median, so capital is trapped in slow lines while fast lines run thin.

The value case supports a sequenced intervention, not a broad inventory build. Disciplined action on the ranked priorities supports an estimated €1.6M twelve-month value pool: €1.5M from recovered lost-sales margin and €106k from released working capital. Nearly half of that pool sits in the Health category and 68% sits in Health and Pet Care together. The six recommendations in Section 8 sequence the work from supplier remediation to inventory rebalancing, ordered by return and dependency.

> **Takeaway** **Fix two suppliers and one product family first. They carry 79% of lost-sales value and the largest single SKU opportunity, and neither fix needs new inventory investment.**

SECTION 2

Context and objectives

The study covers a multi-warehouse fast-moving consumer goods network spanning Iberia and southern France. Four distribution centres serve demand across eight product categories sourced from 12 suppliers. The observation window runs 731 days, from January 2024 to December 2025. Daily records capture demand, fulfilment, lost sales, on-hand and on-order inventory, days of supply, and supplier delivery performance at the product-warehouse grain, producing roughly 351,000 daily rows.

Service performance and inventory exposure pull against each other. Holding more stock lifts the fill rate but ties up working capital and raises obsolescence risk; holding less frees capital but exposes the network to stockouts and lost margin. A network can post an acceptable average and still be failing badly in the places that matter, because good locations mask weak ones and slow lines mask fast ones. The objective is to find the specific points where that balance is failing and to size the value of fixing them.

The analysis answers six questions:

- Where is service failing, and is the trend improving or deteriorating?
- Which suppliers carry the largest downstream lost-sales exposure?
- Where is inventory misaligned with demand rather than simply high?
- How concentrated are the losses across products, locations, and segments?
- What is the recoverable value, and how should the work be sequenced?
- Which assumptions must be validated before the value case is treated as budget-ready?

Network at a glance

Dimension	Coverage
Warehouses	4 (Madrid, Lyon, Porto, Lisbon)
Regions	Spain Central, France South-East, Portugal North and South
Products	120 across 8 categories
Suppliers	12
Product-warehouse pairs scored	480
Observation window	731 days (Jan 2024 - Dec 2025)
Daily fact rows	~351,000
Observed lost-sales value	€20.3M
Observed lost-sales margin (proxy)	€8.8M
Average inventory value (daily)	€4.1M

Companion dashboard

This report has an interactive companion. The published dashboard exposes the same service, supplier, warehouse, and SKU-risk views with filtering down to the product-warehouse pair, so a reader can move from a finding here to the underlying records without rerunning the pipeline. The static charts in this report are the fixed, citable version of those views. Where a figure caption names a processed table, the dashboard reads the same table, and a reconciliation gate enforces that the two never disagree.

SECTION 3

Data and methodology

The pipeline moves from reproducible synthetic data to SQL analytical views, policy-based risk scoring, directional impact estimates, and a published dashboard. Every stage is covered by data contracts and reconciliation gates that block release on any failure or high-severity warning. The sections below describe each metric so that the findings can be read against a precise definition rather than an intuition.

Service and inventory metrics

Fill rate is units fulfilled divided by units demanded. Stockout rate is units of unmet demand divided by units demanded. Days of supply is available inventory divided by average daily demand. Lost-sales value prices unmet demand at unit selling price, and the lost-sales margin proxy applies category gross-margin assumptions to that value. Excess inventory is value held above the ABC-class days-of-supply policy cap, and the slow-moving proxy isolates value sitting on days with no off-take.

Risk scoring

Each product-warehouse pair, supplier, and category-region segment receives a composite governance priority score from 0 to 100, built from five weighted components: service risk, stockout risk, excess inventory, supplier risk, and working-capital risk. Each component is normalized against fixed policy thresholds before weighting, so the score expresses priority against declared operating tolerances. Pairs are tiered Critical, High, Medium, or Low. The score records which of the five components is the main driver for each entity, which is what lets the findings separate a stockout problem from an overstock problem.

Score component	What it captures
Service risk	Fill-rate gap against demand at the entity grain.
Stockout risk	Frequency and depth of unmet-demand days.
Excess inventory	Value held above the ABC days-of-supply policy cap.
Supplier risk	Lead-time variability and on-time delivery shortfall.
Working-capital risk	Capital tied in slow-moving and overstocked lines.

Takeaway Scores prioritise; they do not prove causation. The score points to where to look first, and the underlying component then says why.

Impact and opportunity estimates

Financial figures are directional proxy estimates, not audited profit and loss. Observed values are annualised using a 365-over-731-day factor. The twelve-month value pool combines recoverable lost-sales margin under a scenario recovery rate with releasable trapped working capital. The margin component applies category gross-margin assumptions to the units a scenario recovery rate would convert; the working-capital component applies the same recovery logic to the average trapped capital proxy of €426k per day. Because recovery is partial by design, the pool of €1.6M is a fraction of the €20.3M gross loss, not a claim that all of it is reachable.

Quality controls

Data contracts cover required columns, grain, nulls, ranges, domains, and key references. SQL and Python checks reconcile service, inventory, impact, scoring, and dashboard metrics against each other so that a number shown on the dashboard matches the number in the underlying table. Continuous integration runs the full pipeline, unit tests, release gates, and a dashboard freshness check on every change. The release gate blocks publication on any failure or high-severity warning.

SECTION 4

Analytical framework

The analysis works through four lenses, each answering a different question about the same loss. Reading them in order moves from symptom to cause to size, which is also the order in which the recommendations act.

1. Trend: is the problem getting worse?

A point-in-time fill rate cannot tell remediation from drift. The trend lens tracks fill rate, stockout rate, and lost-sales value month by month and compares the first and last 90-day windows by location. A declining baseline changes the economics of any fix, because inventory added to a sliding system is absorbed rather than converted to service.

2. Concentration: where is the loss, exactly?

Averages hide the structure that matters for action. The concentration lens ranks SKU-location pairs, categories, ABC classes, and category-region segments by their share of loss, then measures how much of the total sits in the worst few. Sharp concentration means a short list of targeted fixes will outperform a uniform service target applied across the catalogue.

3. Cause: supply reliability versus inventory placement

Lost sales come from two distinct failures: stock that never arrives reliably, and stock that sits in the wrong place. The cause lens separates them by reading supplier on-time performance and lead-time variability against warehouse fill rate and days of supply. The two failures need different fixes, so labelling each loss correctly is what keeps the plan from over-investing in inventory.

4. Size: what is recoverable, and in what order?

The final lens converts the diagnosis into a ranked value pool. It estimates recoverable margin and releasable working capital under a scenario recovery rate, attributes the pool to categories, suppliers, warehouses, and SKUs, and orders the work by return and dependency. The output is a sequence, not a wish list, because the first fixes are preconditions for the later ones to pay back.

> **Takeaway** **Symptom, structure, cause, size. Each finding section below maps to one of these lenses, and the recommendations follow the same order.**

SECTION 5

Decision case

The evidence supports three management decisions. First, the near-term service plan should be funded as a supplier and SKU recovery effort, not as a network-wide safety-stock increase. Second, the first inventory move should be a reallocation from slow lines and stronger sites toward the underserved Madrid and Lyon lanes. Third, the value case should be managed against recoverable margin first and working-capital release second, because the financial loss is caused primarily by missed demand.

The decision logic below translates the analytical readout into action. It separates the operating hypothesis, the evidence already strong enough to act on, and the validation required before committing material capital. This is the difference between a diagnostic report and an execution case: the next step is not more broad analysis, but controlled remediation with specific proof points.

Supplier remediation. Supplier 2 and Supplier 3 carry 79% of lost-sales value; their on-time rate is 72% versus 90% for the rest of the base. Prioritise lead-time stability, constrained-lane expediting, and supplier-specific recovery governance before adding stock.

SKU recovery. All 4 High-tier SKU-location pairs are Health Product 16, worth €530k of ranked SKU-location value. Run one cross-warehouse recovery plan covering supply source, allocation rules, reorder point, and substitution risk.

Inventory placement. Madrid and Lyon generate 62% of warehouse lost sales while Lisbon and Porto hold 59% of inventory value. Move stock selectively after supplier inputs stabilise; avoid a blanket stock build that would raise capital without fixing drift.

Category focus. Health and Pet Care hold 68% of the recoverable value pool; Health alone is 47% of lost sales at 93.3% fill. Create a Health and Pet Care service cell with weekly category-level decisions rather than diffuse catalogue-wide reviews.

What would change the decision

The current evidence is sufficient to start the no-regret actions: supplier stabilisation, Health Product 16 recovery, and SKU-level policy review. Three findings would change the investment case before material inventory is moved. The first is a commercial substitution effect showing that a large share of stockout demand was captured by another SKU rather than lost. The second is a supplier contract or promotion calendar proving that Health Product 16 demand was abnormal and non-repeatable. The third is a warehouse capacity constraint that prevents Madrid or Lyon from absorbing rebalanced stock without higher handling cost. None invalidates the diagnosis; each changes the size, sequencing, or net economics of the fix.

> Takeaway **Approve the first wave as a controlled recovery programme. Hold back broad inventory investment until substitution, promotion, and capacity checks confirm the net value case.**

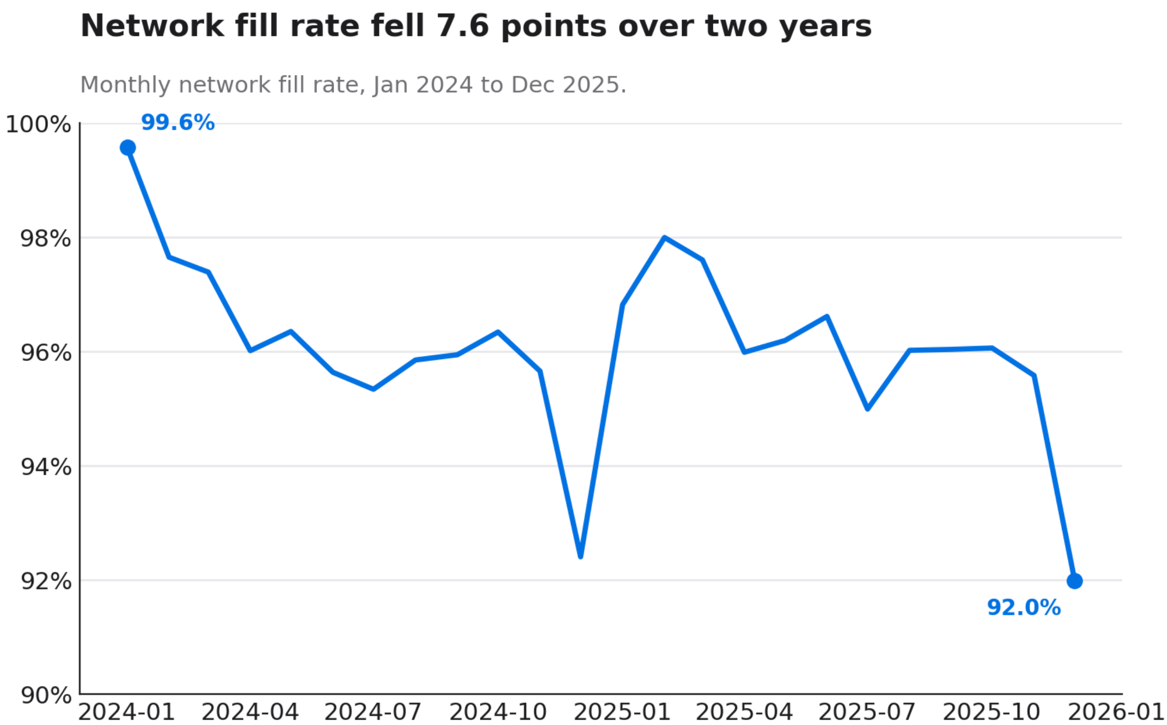
SECTION 6

Findings

The findings are grouped into six themes. The first establishes that service is deteriorating, the next three locate the loss across suppliers, warehouses, and the catalogue, the fifth measures how concentrated it is, and the last sizes the recoverable pool.

6.1 Service quality is on a two-year decline

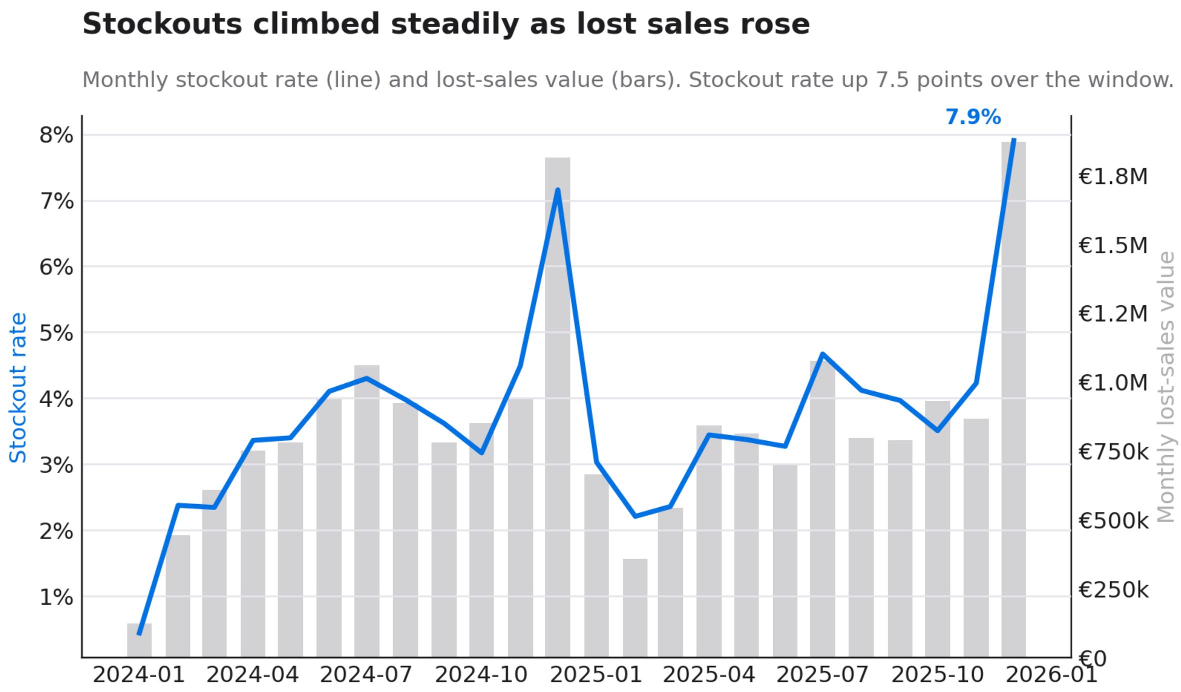
The monthly network fill rate fell from 99.6% in January 2024 to 92.0% in December 2025. The path is not a one-off shock. The rate drifts down through 2024, recovers partially in early 2025, then slides again to its lowest point in the final month of the window. December 2025 is the worst month in the series on both fill rate and lost-sales value, which means the network entered 2026 at its weakest observed state rather than recovering toward the mean.



Source: data/processed/daily_product_warehouse_metrics.csv. Fill rate = units fulfilled / units demanded.

Figure 1. Monthly network fill rate, Jan 2024 to Dec 2025.

Reading the stockout rate alongside lost-sales value confirms the same story from the cost side. The stockout rate climbed from 0.4% to 7.9% over the window, and monthly lost-sales value rose with it, peaking in the final months. The last three months averaged 3.1x the lost-sales value of the first three months, so the end-state is not just lower service; it is a materially more expensive operating baseline. A declining baseline matters for planning: inventory added without fixing replenishment discipline will be absorbed by the drift rather than converted into service, so stabilising the trend is a precondition for any inventory investment to pay back.



Source: data/processed/daily_product_warehouse_metrics.csv. Stockout rate = share of product-warehouse-days with unmet demand.

Figure 2. Monthly stockout rate (line) against lost-sales value (bars).

The decline is general, not local. Comparing the first and last 90 days of the window, every warehouse serves worse at the end than at the start. Madrid falls 5.4 points and Lyon 4.2, the two that were already weakest, while even the strongest sites, Lisbon and Porto, give up 2.3 and 3.7 points. A decline that touches every location points to a system-level cause in replenishment rather than a problem isolated to one hub.

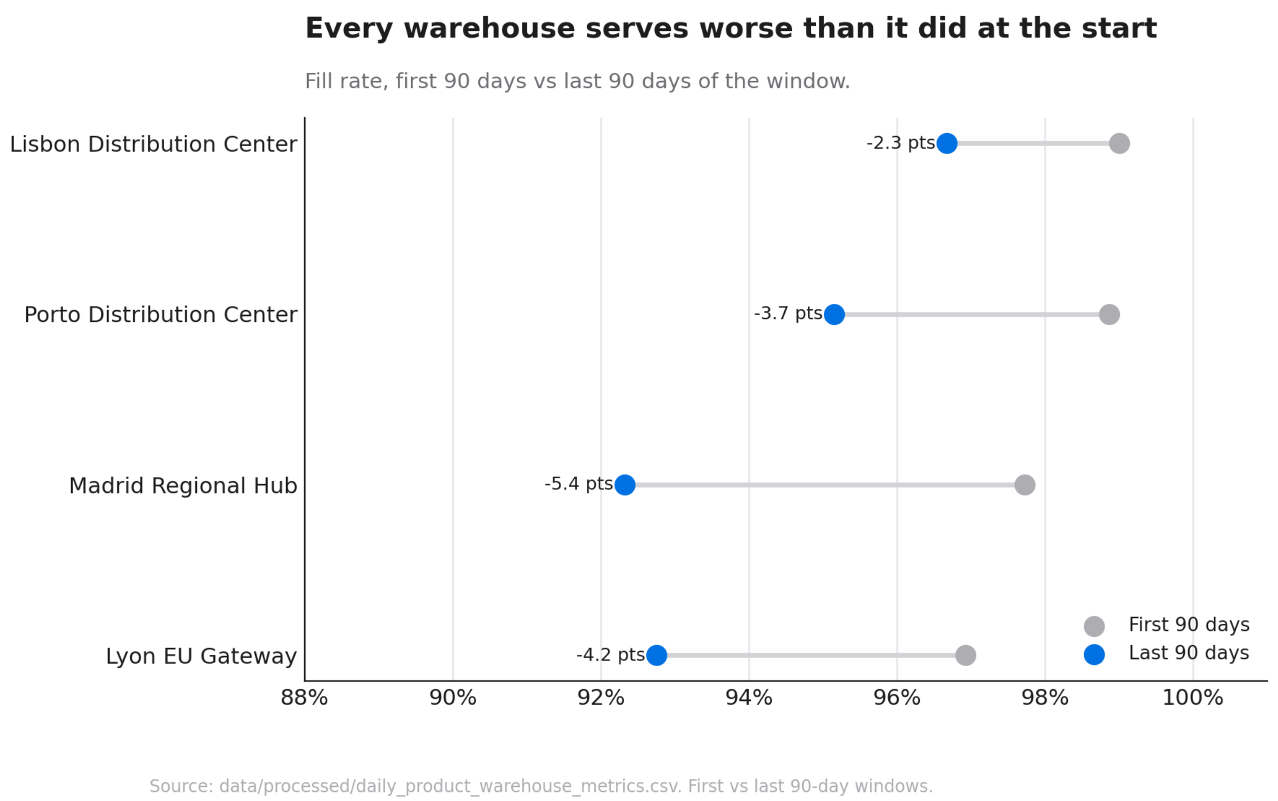
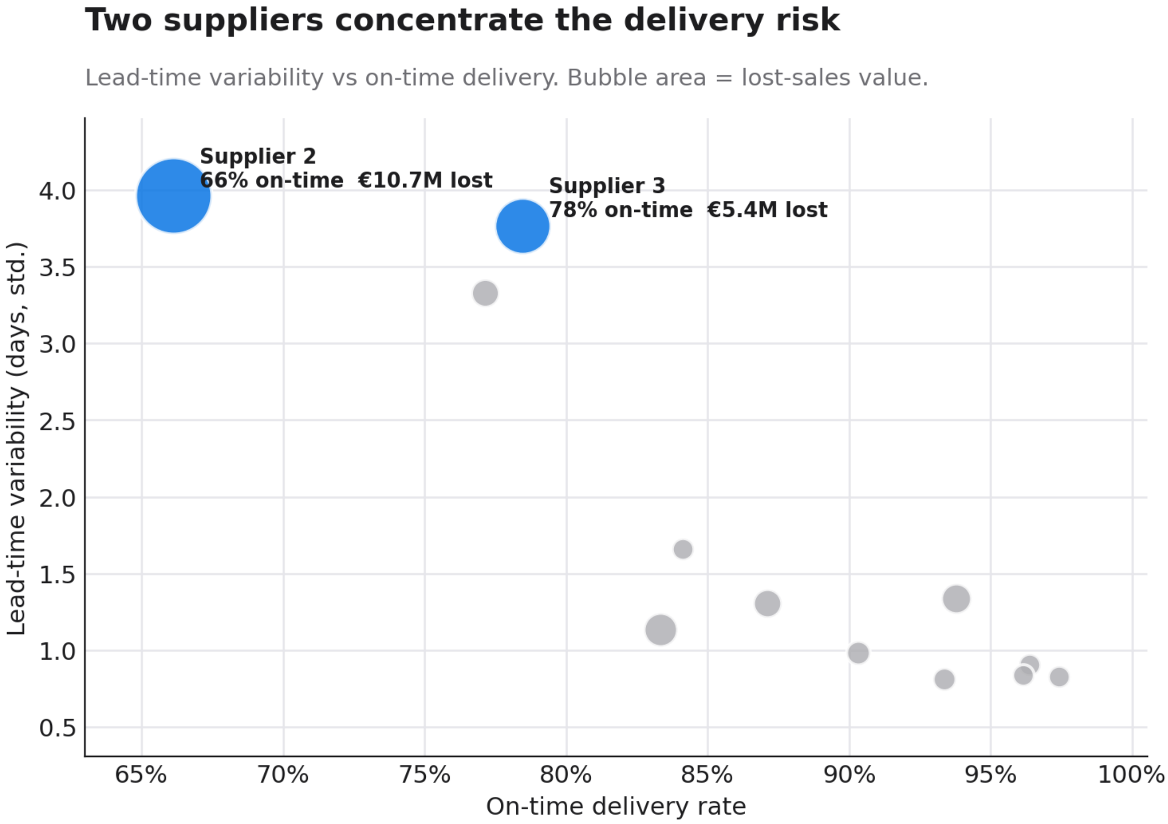


Figure 3. Fill rate, first 90 days versus last 90 days, by warehouse.

> **Takeaway** **Treat the decline as structural and network-wide. Stabilising replenishment is the first move, before any inventory is added.**

6.2 Two suppliers carry four fifths of the exposure

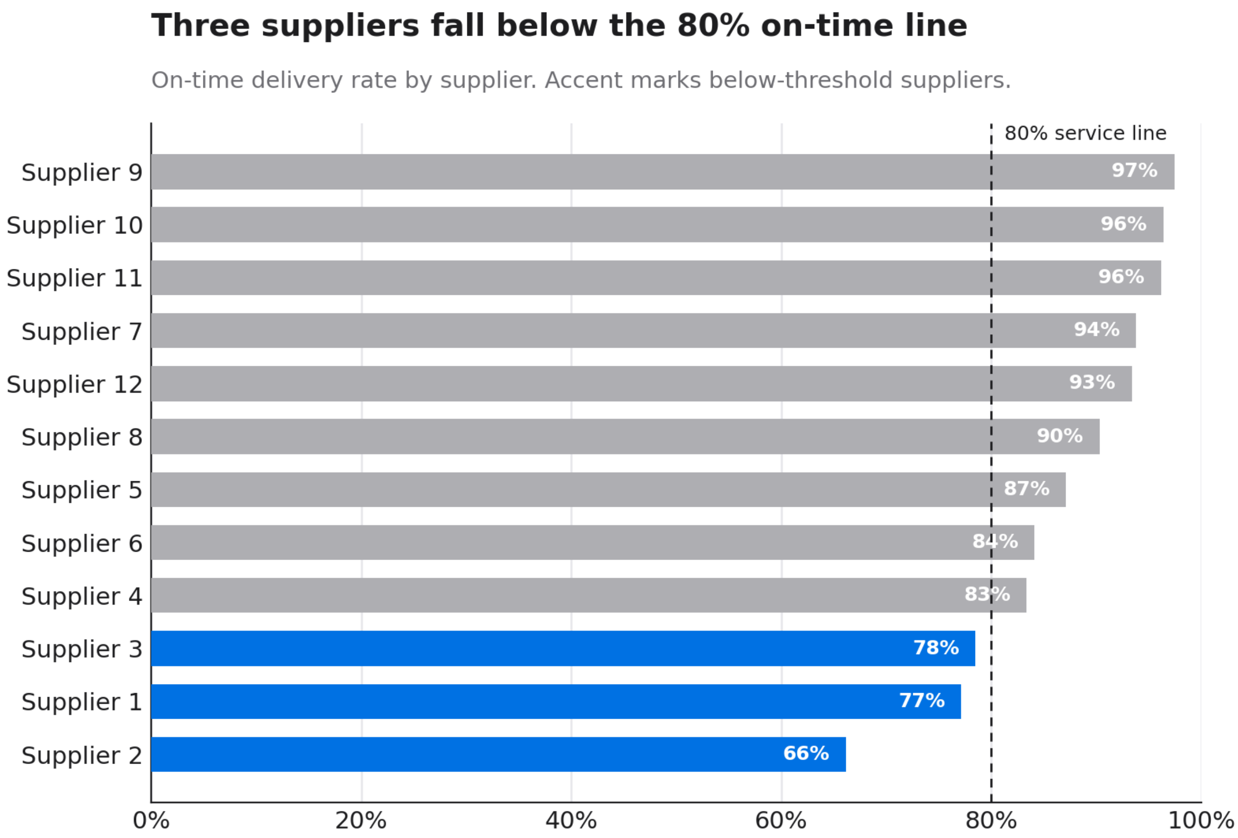
Supplier 2 and Supplier 3 are the only two suppliers in the High risk tier. Supplier 2 delivers on time on just 66% of orders with lead-time variability of 4.0 days; Supplier 3 sits at 78% on-time and 3.8 days. Together they are associated with €16.0M of lost-sales value, 79% of the network total. The remaining ten suppliers cluster above 83% on-time with variability below 1.7 days. The portfolio-level contrast is material: high-risk suppliers average 13.0% stockout exposure versus 1.4% for the rest. The relationship is clear enough for action: unreliable lead times, not low received quantity, drive the lost sales, since both high-risk suppliers still ship more than 92% of ordered units once they arrive.



Source: data/processed/supplier_risk_table.csv. Bubble area scaled to annualized lost-sales value associated with each supplier.

Figure 4. Lead-time variability against on-time delivery. Bubble area is lost-sales value.

Ranked on on-time delivery alone, three suppliers fall below an 80% service line: Supplier 2 at 66%, Supplier 1 and Supplier 3 in the high 70s. Supplier 1 sits in the Medium tier because the volume behind it is smaller, so it carries less lost-sales value despite similar reliability. This is why the priority order follows exposure, not the reliability percentage on its own: the same delay rate matters more behind a high-volume lane than a thin one.



Source: data/processed/supplier_performance_summary.csv. On-time rate = orders received on or before due date.

Figure 5. On-time delivery rate by supplier against the 80% service line.

The practical test is simple. If lead-time variability falls on the two high-risk suppliers and the affected SKUs recover before additional inventory is deployed, the supplier hypothesis is confirmed. If service does not recover, the next diagnostic branch is allocation policy, order sizing, and demand volatility. That validation path keeps the response disciplined: fix the highest-probability cause first, but instrument the result so the organisation does not mistake correlation for proof.

> **Takeaway** **Supplier 2 and Supplier 3 are the single highest-return intervention. Expedite constrained lanes and renegotiate lead-time variability before any broad safety-stock increase.**

6.3 Inventory is misaligned, not uniformly excessive

Madrid and Lyon post the network's worst fill rates, 94.6% and 94.1%, while holding the least inventory value, around €927k and €764k. Lisbon and Porto carry the most inventory, €1.4M and €1.0M, and deliver the best service at 97.8% and 96.9%. The fill-rate spread of 3.7 points across warehouses tracks inventory placement, not total volume. Stock is sitting where service is already strong and is thin where it is failing.

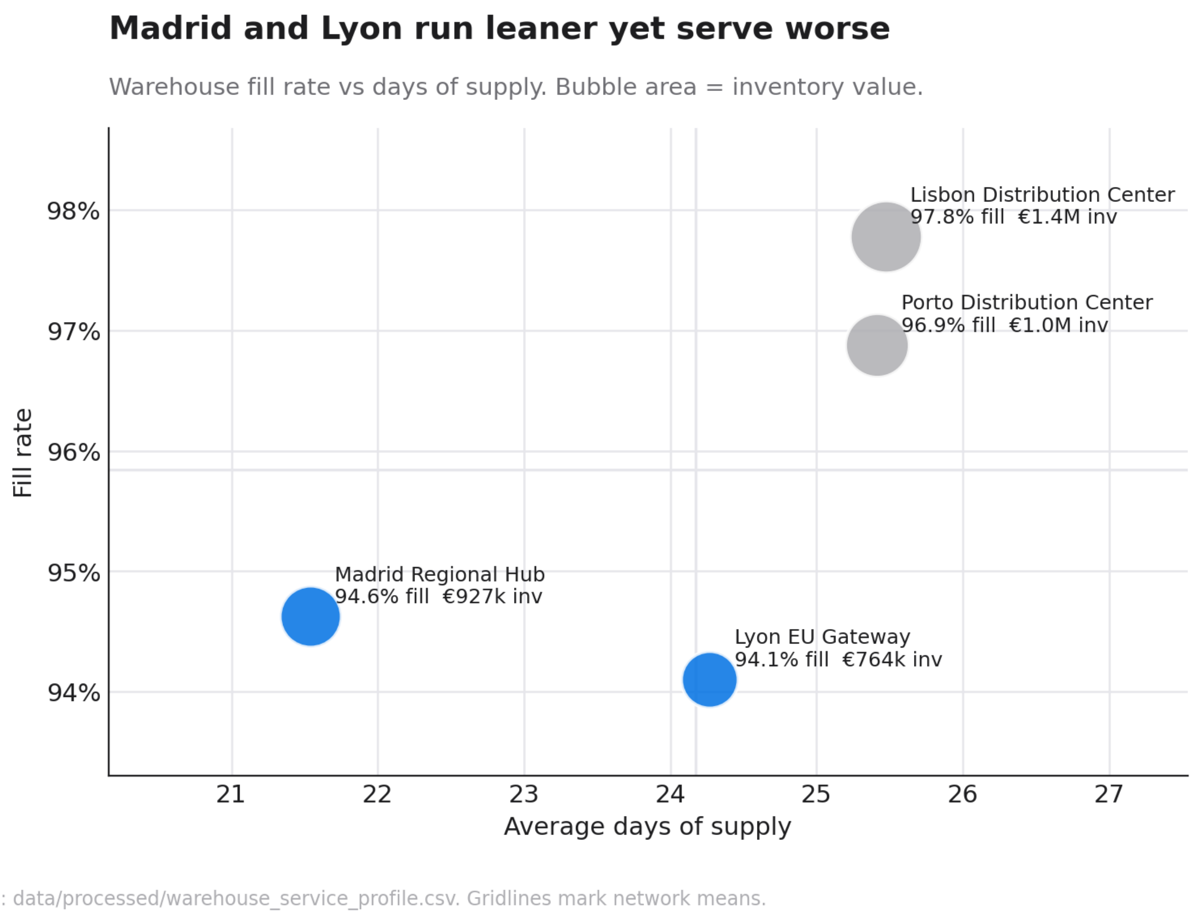
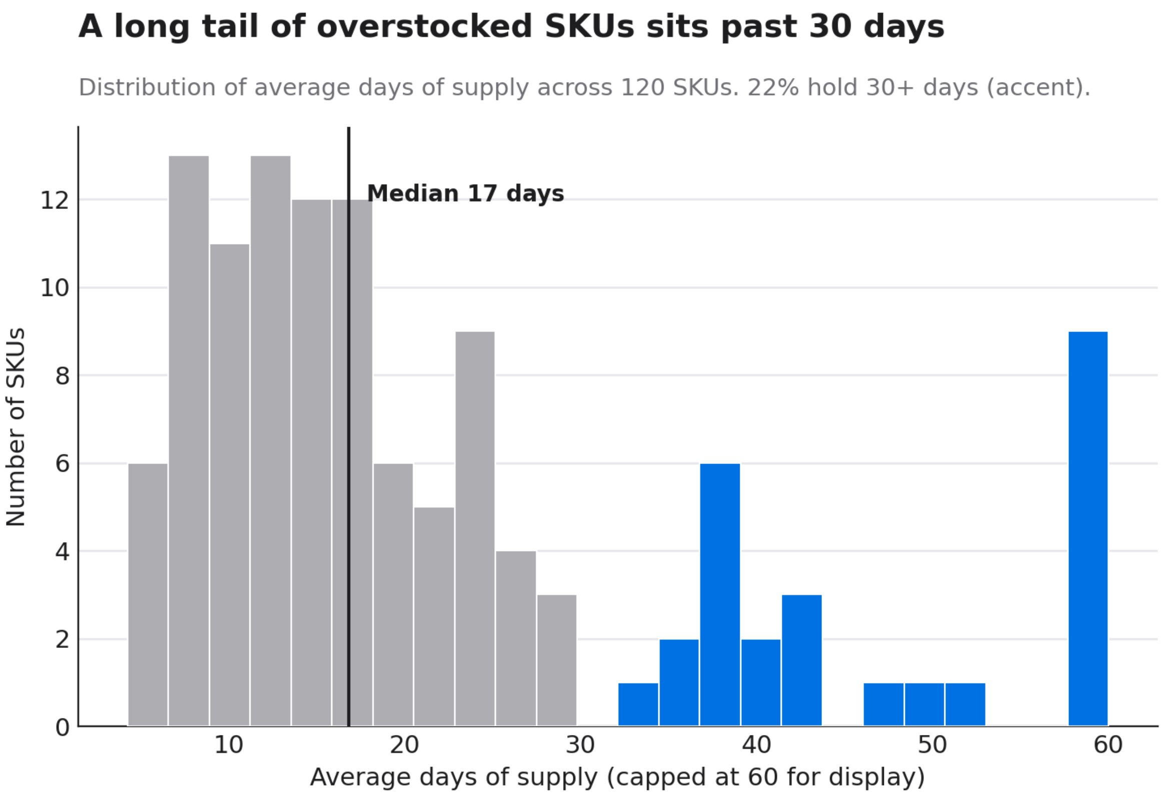


Figure 6. Warehouse fill rate against days of supply. Bubble area is inventory value.

The same misalignment shows at SKU level. The median SKU holds 17 days of supply, but 22% of SKUs carry 30 days or more, and a thin tail stretches past 231 days. Capital is trapped in slow lines, an average of €426k per day in the trapped working-capital proxy and €410k per day above the ABC days-of-supply caps, while the fast lines that drive lost sales run short. The problem is distribution and policy, not a uniform shortage or a uniform glut.



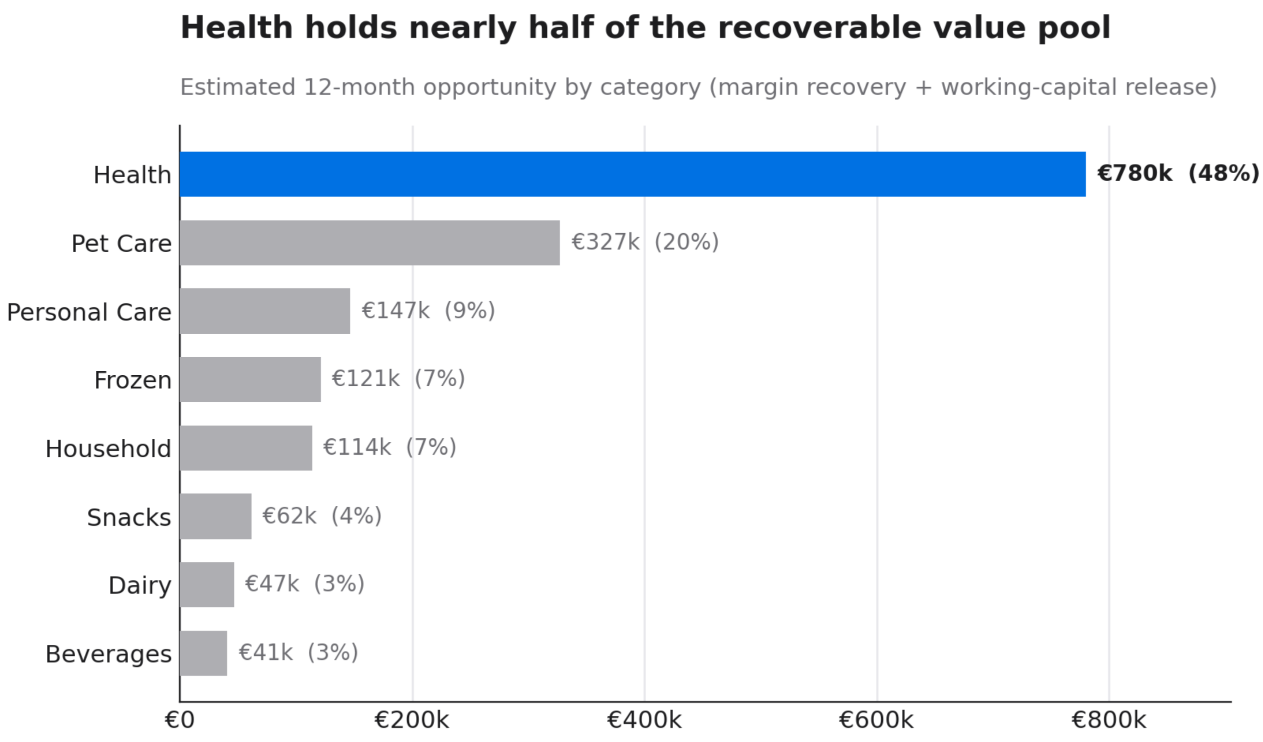
Source: data/processed/product_inventory_profile.csv. Values capped at 60 days for display; a thin tail extends to 231 days.

Figure 7. Distribution of average days of supply across 120 SKUs (capped at 60 for display).

Takeaway **Rebalance toward Madrid and Lyon and trim the 30-plus-day tail rather than adding network-wide stock. The service gap is a placement gap.**

6.4 Half the value pool sits in one category

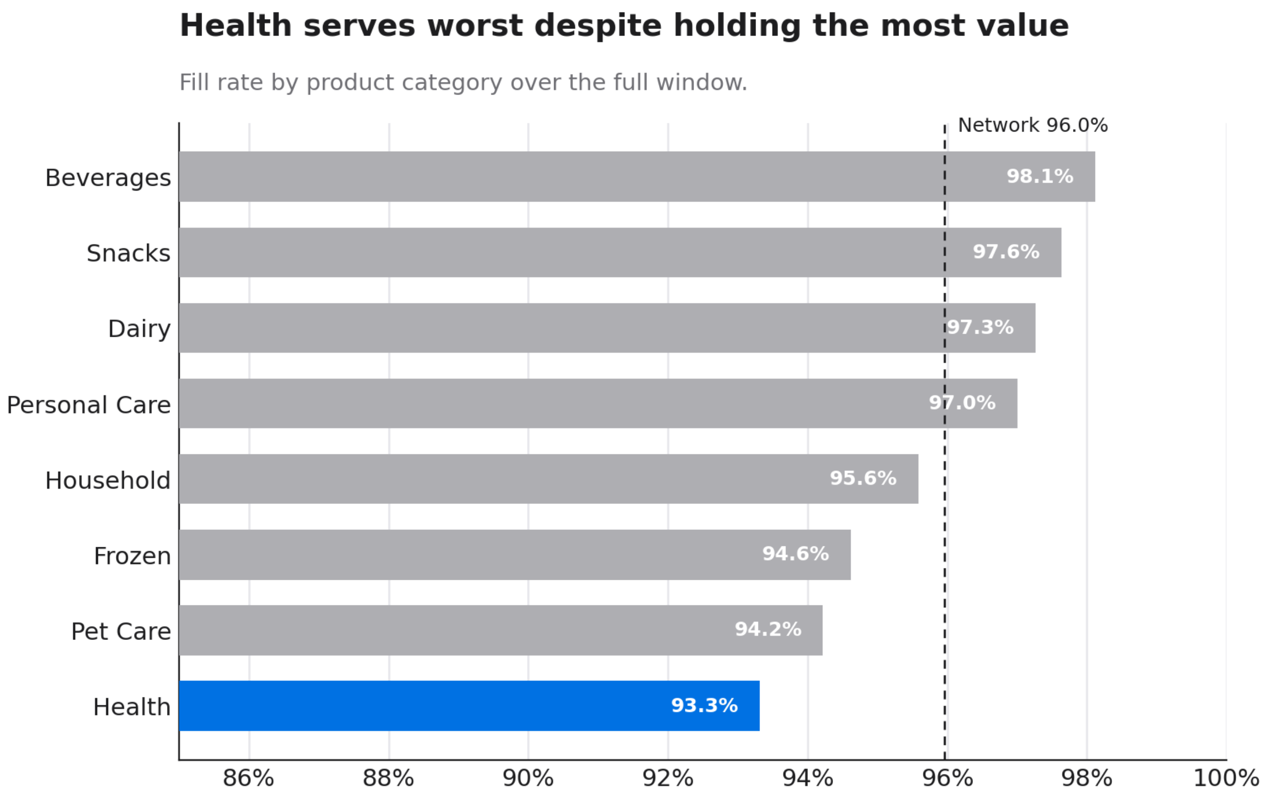
The estimated twelve-month opportunity totals €1.6M. The Health category alone holds €780k of it, 48% of the pool, with Pet Care second at €327k (20%). The remaining six categories split the final third. Concentration this sharp means a category-led operating rhythm will outperform a uniform service target applied across the catalogue.



Source: outputs/tables/impact_opportunity_priority.csv. Proxy 12-month value pool under the scenario recovery rate.

Figure 8. Estimated 12-month opportunity by category.

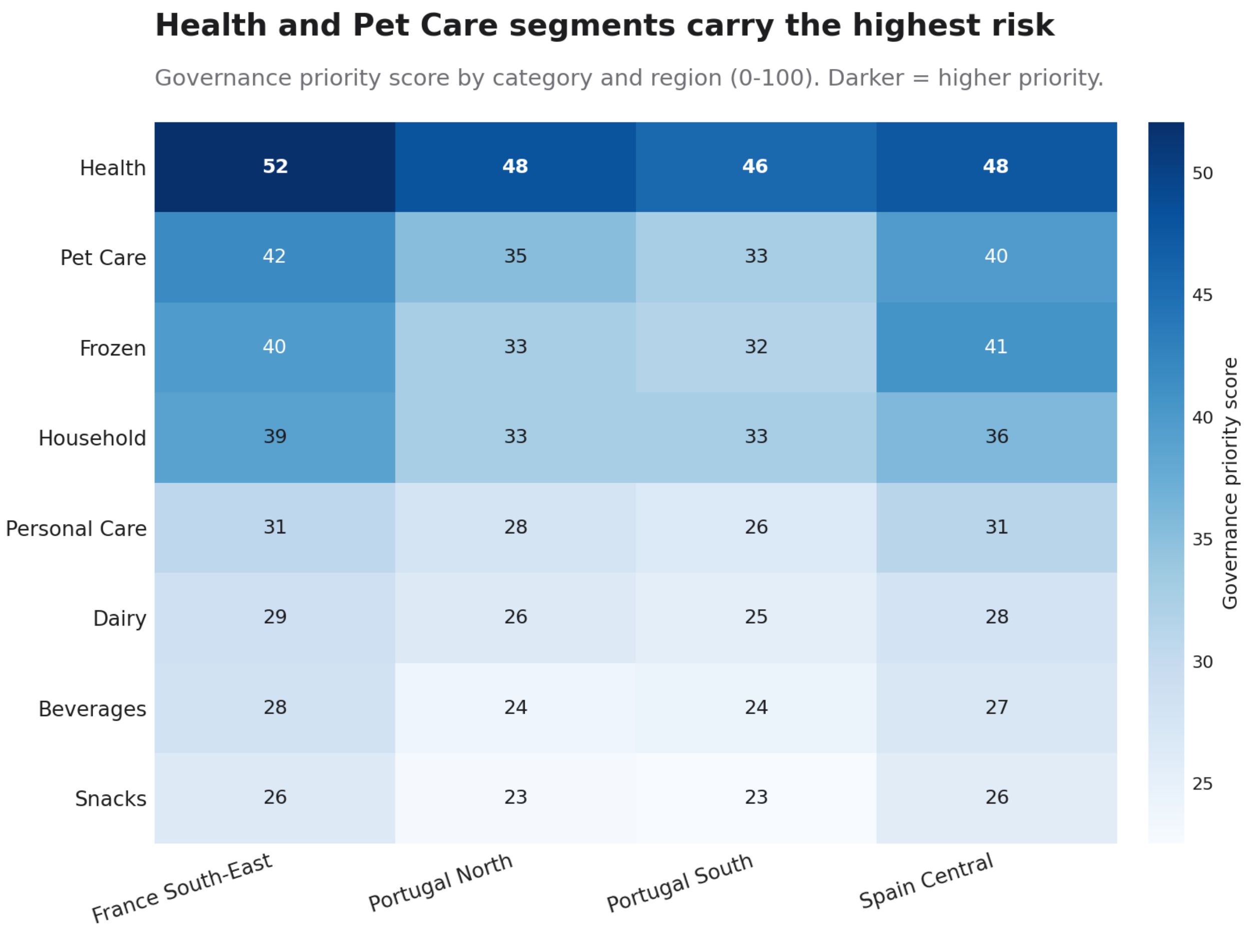
Health does not lead the pool because it is large; it leads because it serves worst. Health posts the lowest fill rate of any category at 93.3% and the highest lost-sales value at €9.60M, despite holding among the most inventory value. It represents 47% of lost sales, while Pet Care adds another 22%. Pet Care is second worst on both. The categories that serve well, Beverages at 98.1% and Snacks at 97.6%, contribute almost nothing to the pool. Service quality, not category size, sets the priority order.



Source: data/processed/daily_product_warehouse_metrics.csv. Fill rate aggregated over 24 months.

Figure 9. Fill rate by product category over the full window.

The category signal repeats at the category-region grain. Every Health segment scores in the high 40s to low 50s on governance priority, the highest band in the network, and each one is driven by stockout risk rather than overstock. Health in France South-East tops the segment ranking at 52. Pet Care segments form the next band. No region escapes the Health problem, which reinforces that the fix belongs at category and product level rather than at a single site.



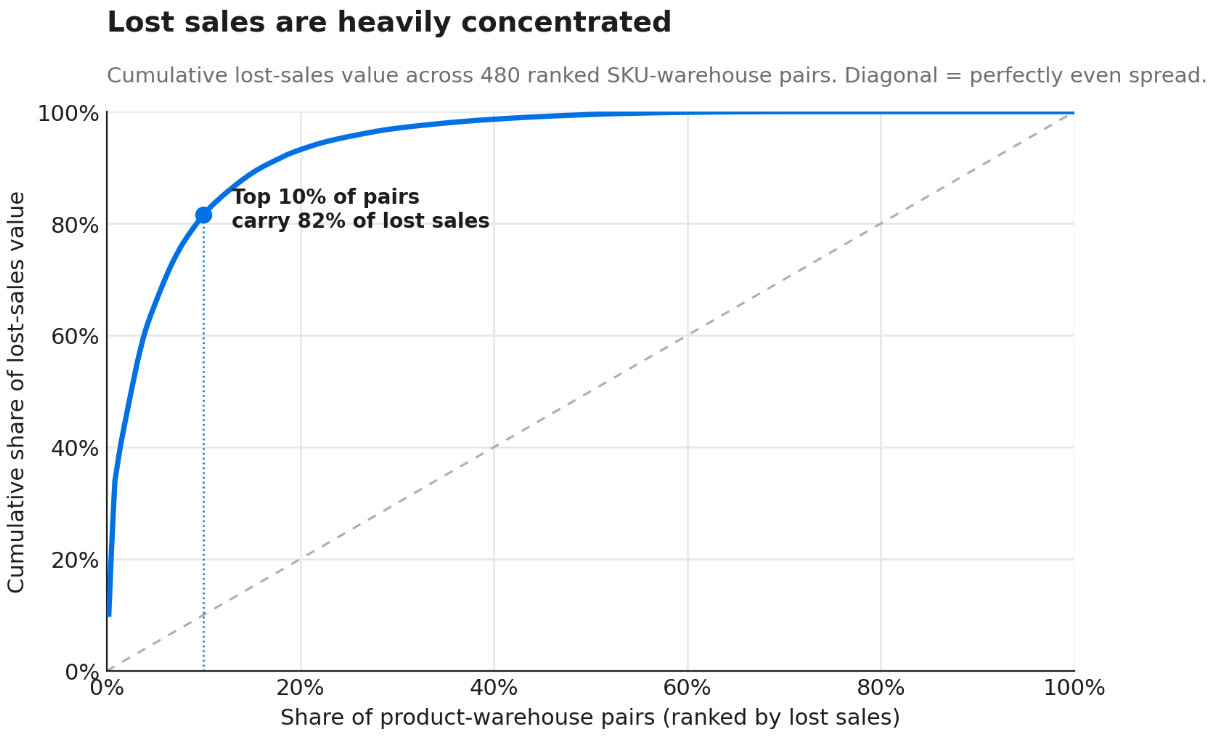
Source: data/processed/segment_risk_table.csv. Composite governance priority score by category-region segment.

Figure 10. Governance priority score by category and region (0-100).

Takeaway **Stand up a Health-category service cell first. Health and Pet Care together hold 68% of the recoverable value.**

6.5 The loss is concentrated in a few SKU-location pairs

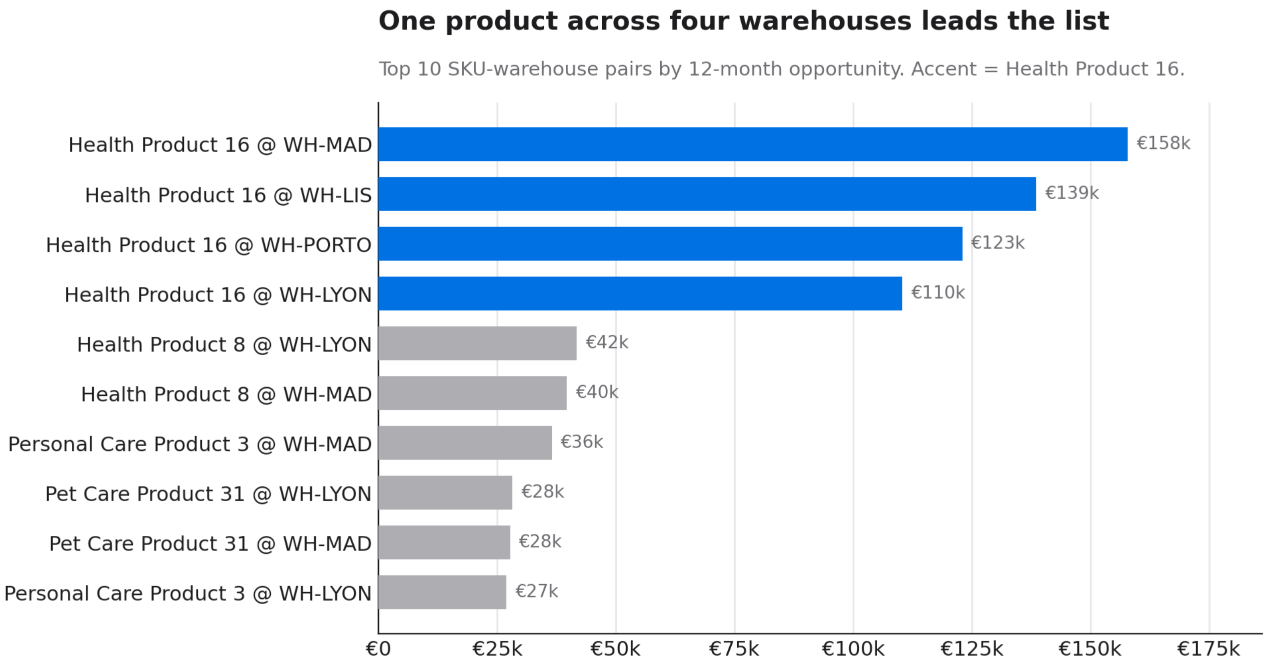
Of the 480 product-warehouse pairs scored, only 4 reach the High tier and 57 reach Medium; the other 419 are Low. The lost-sales value behind them is far more skewed than the tier counts suggest. The worst 10% of pairs carry 82% of all lost sales and the worst 5% carry 66%. A short, well-chosen list of interventions therefore reaches most of the loss without touching the long tail of pairs that lose almost nothing.



Source: data/processed/sku_risk_table.csv. Lorenz-style cumulative concentration curve.

Figure 11. Cumulative share of lost-sales value across 480 ranked SKU-warehouse pairs.

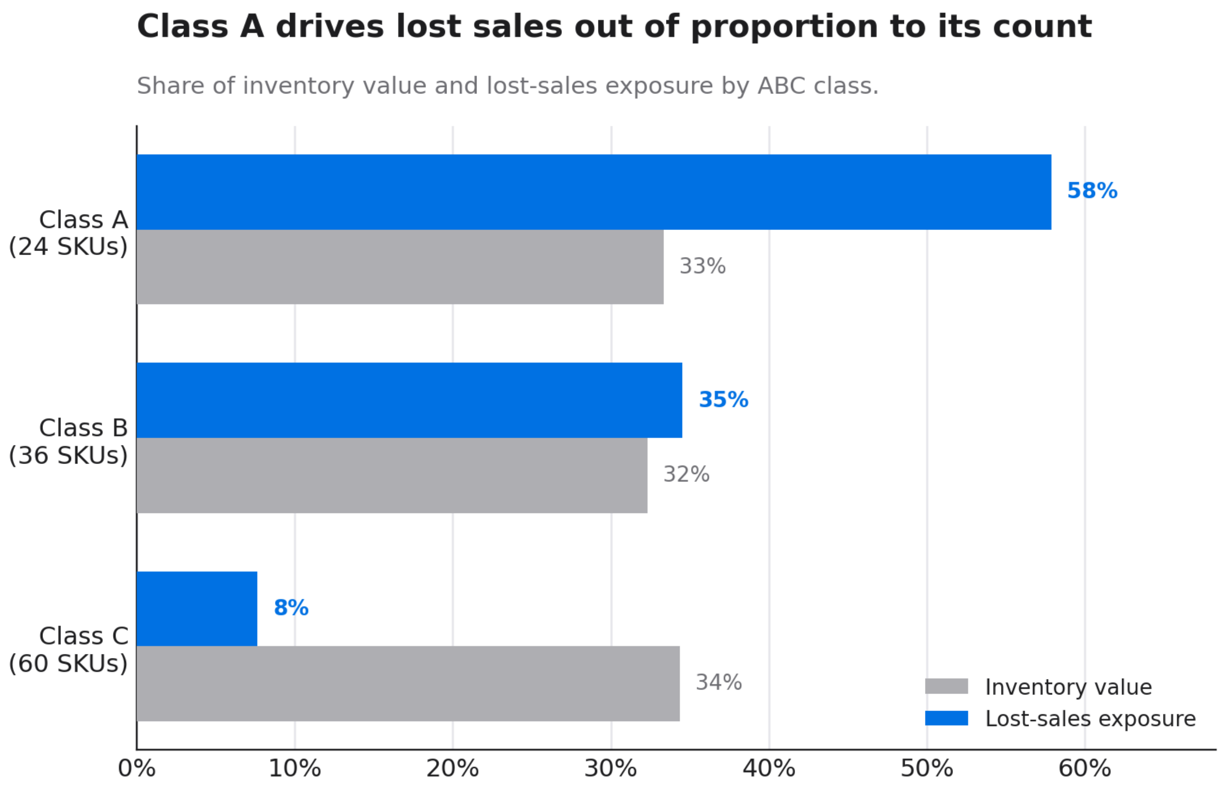
All four High-tier pairs are the same product, Health Product 16, one in each warehouse. Combined they represent €530k of twelve-month opportunity and 62% of the ranked SKU-location pool. The single largest pair is Health Product 16 in Madrid at roughly €158k. A product that fails in all four warehouses at once is failing for a product-level reason, supply or planning, not a local one, so it should be run as one cross-warehouse recovery rather than four separate fixes.



Source: outputs/tables/impact_opportunity_priority.csv. SKU-warehouse grain, proxy 12-month value.

Figure 12. Top 10 SKU-warehouse pairs by 12-month opportunity.

The ABC view explains why so few lines carry so much. Class A holds 24 of the 120 SKUs and a third of inventory value, but 58% of lost-sales exposure. Class C, 60 SKUs and a third of inventory value, carries only 8% of the loss while absorbing capital that the fast lines need. The network is overstocked on slow C lines and underserved on fast A lines. That is a policy and placement gap, not evidence that the network needs a larger inventory budget.



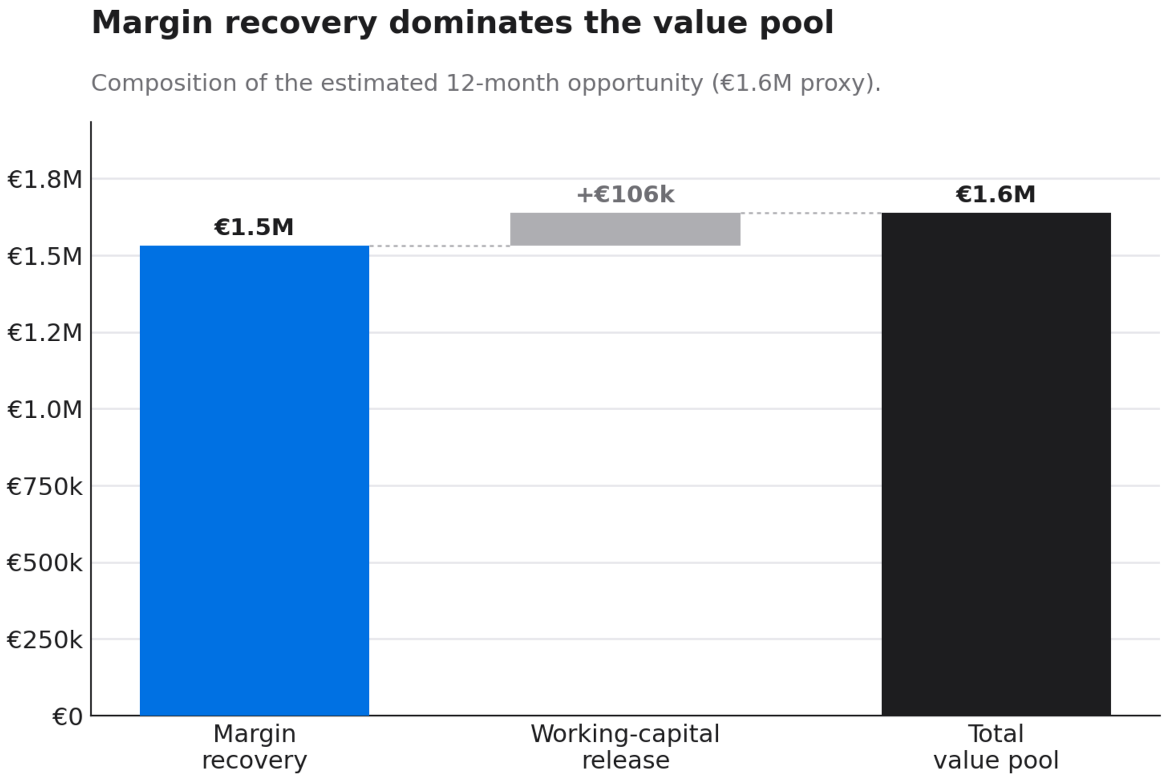
Source: data/processed/product_inventory_profile.csv. ABC class by average inventory value.

Figure 13. Share of inventory value and lost-sales exposure by ABC class.

> **Takeaway** **Run a dedicated recovery plan for Health Product 16 across all four warehouses and reweight the ABC policy from C toward A. The loss lives in a small, addressable set of fast lines.**

6.6 The recoverable pool is real but partial

The twelve-month value pool of €1.6M splits into €1.5M of recoverable lost-sales margin and €106k of releasable working capital. Margin recovery dominates because the network's core problem is service, not capital efficiency: the gross loss to stockouts of €20.3M is large, and even a partial recovery of its margin outweighs the capital that can be safely released from overstocked lines. The pool is deliberately a fraction of the gross loss, because the scenario recovery rate assumes only part of unmet demand can be converted once service is restored.



Source: outputs/tables/impact_overall_summary.csv. Proxy estimates: recoverable lost-sales margin plus releasable working capital.

Figure 14. Composition of the estimated 12-month opportunity (proxy).

The working-capital component is modest by design. The trapped-capital proxy averages €426k per day, but only a portion is releasable without re-exposing service, so the pool credits €106k rather than the full balance. This is the right asymmetry for a network whose losses come from missing the sale, not from carrying too much: the prize is protecting and recovering demand, with capital release as a secondary benefit from trimming the slow-moving tail.

> **Takeaway** **Manage to the margin pool first. Working-capital release is a by-product of fixing service and trimming slow lines, not a separate programme.**

SECTION 7

Risks, limitations, and caveats

The findings are only as strong as the assumptions behind them. This section states the main limits plainly so that the recommendations are read with the right level of confidence.

Data scope

The dataset is synthetic and does not represent a specific company. It is constructed to exhibit the structural patterns common in multi-warehouse FMCG operations: a service decline, supplier concentration, and ABC inventory skew. The findings describe the modelled network; the methods apply directly to operational data.

Proxy financials, not audited profit and loss

Every euro figure is a directional proxy. Lost-sales value prices unmet demand at list price and assumes the demand was real and would have converted. The margin proxy applies category-level gross-margin assumptions rather than SKU-level actuals. The opportunity pool applies a scenario recovery rate. None of these are audited profit and loss, and the absolute numbers would move with different assumptions. The relative structure, which suppliers, products, and locations dominate, is far more stable than the absolute totals.

Correlation, not proven causation

The governance priority score ranks where to look; it does not prove why a line fails. The supplier finding rests on a strong association between lead-time variability and lost sales, but the data cannot rule out a confound such as those suppliers serving inherently harder demand. The Health Product 16 finding identifies a product that fails everywhere, which is suggestive of a product-level cause, but confirming it requires operational review of the supply and planning records behind that SKU.

Scope boundaries

The analysis covers service and inventory outcomes. It does not model price elasticity, promotional cannibalisation, substitution between SKUs when one stocks out, or the cost of the interventions themselves. A stockout on one product may shift demand to another rather than lose it outright, which would lower the true lost-sales figure. The recommended actions carry implementation cost that is not netted against the value pool. Both effects argue for treating the pool as an upper-bound on service-side value rather than a net benefit.

> **Takeaway** **Trust the structure more than the totals. The ranking of suppliers, products, and locations is more stable; the absolute euro figures are scenario estimates that move with the assumptions.**

SECTION 8

Recommendations and action priorities

The network's service problem is concentrated enough to act on precisely. Two suppliers, one product family, and two warehouses form a concentrated intervention set within the €20.3M lost-sales exposure. The declining fill-rate trend means the window to act is closing, but it also means targeted fixes will show up clearly against the baseline. The six recommendations below are ordered by return and by dependency: the first two carry the bulk of the value and require no inventory investment.

Recommendation	Horizon	Priority
1. Remediate Supplier 2 and Supplier 3 Expedite replenishment on their constrained lanes and renegotiate lead-time variability toward the sub-1.7-day network norm. Addresses 79% of lost-sales value with no added stock.	0-3 months	Highest
2. Launch a Health Product 16 recovery plan Treat the four High-tier pairs as one cross-warehouse effort: review the supply source, reset safety stock, and protect allocation. Largest single SKU opportunity at €530k.	0-3 months	High
3. Stand up a Health and Pet Care service cell Own the two categories that hold 68% of the €1.6M pool, with weekly review against the fill-rate baseline and stockout-driven safety-stock rules.	0-6 months	High
4. Rebalance inventory to Madrid and Lyon Shift placement from Lisbon and Porto toward the two underserved hubs rather than adding network-wide stock. Closes the 3.7-point fill spread without raising total inventory.	3-6 months	Medium-High
5. Reweight the ABC days-of-supply policy Trim the 22% of SKUs holding 30-plus days, concentrated in Class C, and redeploy the freed capital to Class A lines that carry 58% of lost-sales exposure.	3-9 months	Medium
6. Instrument the trend and hold the line Track monthly fill and stockout against the December 2025 baseline on the dashboard, with an alert when any warehouse drops below its 90-day average. Prevents the structural drift from resuming after the fixes.	Ongoing	Medium

Sequencing logic

The order is deliberate. Supplier remediation and the Health Product 16 plan come first because they carry the largest return, need no capital, and stabilise the inputs that every later fix depends on. The category service cell institutionalises that focus. Inventory rebalancing and the ABC policy change come next, once supply is reliable enough that moving stock does not simply relocate the stockout. The monitoring step runs throughout. Executed in sequence, the plan targets the €1.6M pool, €1.5M of recoverable margin and €106k of released capital, with the first two actions carrying most of it.

Execution governance

The programme should be governed as a 12-week recovery sprint followed by a six-month operating cadence. The first four weeks should validate the supplier and SKU hypotheses, lock the baseline, and agree exception rules for constrained allocation. Weeks five to twelve should execute supplier stabilisation, Health Product 16 protection, and the first Madrid/Lyon rebalancing moves. The six-month cadence should then institutionalise the category service cell, ABC policy changes, and monitoring rules.

Supplier 2 and 3 recovery. Owner: Procurement / Supply Planning. Target signal: on-time rate moves from 72% toward the 90% non-high supplier benchmark, with lead-time variability trending below 2.0 days.

Health Product 16 protection. Owner: Category / Inventory Planning. Target signal: all four warehouse pairs exit High tier and SKU stockout rate falls before broad category stock is added.

Madrid and Lyon rebalancing. Owner: Network Planning. Target signal: the warehouse fill spread narrows from 3.7 points without raising total network inventory value.

ABC policy reset. Owner: Inventory Governance. Target signal: the 30-plus-day SKU share falls from 22% while Class A lost-sales exposure declines.

Value tracking. Owner: Finance / Analytics. Target signal: recovered margin run-rate tracks toward the €1.5M 12-month proxy, with capital release measured separately from service gain.

Further questions before scale-up

Substitution. What share of stockout demand is truly lost versus substituted to adjacent SKUs, especially inside Health and Pet Care?

Demand abnormality. Did promotions, commercial commitments, or one-off allocation rules distort Health Product 16 demand during the final quarter of 2025?

Warehouse capacity. Do Madrid and Lyon have the physical capacity, labour coverage, and inbound appointment availability to absorb rebalanced stock without creating a new bottleneck?

Supplier constraints. Which supplier terms, minimum-order constraints, or production calendars prevent Supplier 2 and Supplier 3 from reducing lead-time variability?

Net economics. What implementation cost should be netted against the value pool before the programme is converted from recovery sprint to recurring operating model?

> Takeaway **Do the two no-capital fixes first, then prove the inventory move. Supplier discipline and one product recovery plan reach most of the value before any broad stock decision is needed.**

SECTION 9

Appendix

The tables below give the supporting detail behind the findings. All figures are reproducible from the source pipeline.

A. Supplier risk ranking

Supplier	Tier	On-time	LT var (d)	Lost sales	Priority
Supplier 2	High	66%	4.0	€10.7M	63
Supplier 3	High	78%	3.8	€5.4M	63
Supplier 1	Medium	77%	3.3	€663k	39
Supplier 4	Low	83%	1.1	€1.3M	32
Supplier 7	Low	94%	1.3	€900k	31
Supplier 5	Low	87%	1.3	€713k	29
Supplier 6	Low	84%	1.7	€62k	29
Supplier 8	Low	90%	1.0	€242k	26
Supplier 12	Low	93%	0.8	€156k	23
Supplier 10	Low	96%	0.9	€63k	23
Supplier 9	Low	97%	0.8	€59k	21
Supplier 11	Low	96%	0.8	€41k	20

B. Warehouse service and inventory profile

Warehouse	Region	Fill	Stockout	DOS	Inventory
Lyon EU Gateway	France South-East	94.1%	5.9%	24.3	€764k
Madrid Regional Hub	Spain Central	94.6%	5.4%	21.5	€927k
Porto Distribution Center	Portugal North	96.9%	3.1%	25.4	€1.0M
Lisbon Distribution Center	Portugal South	97.8%	2.2%	25.5	€1.4M

C. Twelve-month opportunity by category

Category	Opportunity (12m proxy)	Share	Fill rate
Health	€780k	48%	93.3%
Pet Care	€327k	20%	94.2%
Personal Care	€147k	9%	97.0%
Frozen	€121k	7%	94.6%
Household	€114k	7%	95.6%
Snacks	€62k	4%	97.6%

Category	Opportunity (12m proxy)	Share	Fill rate
Dairy	€47k	3%	97.3%
Beverages	€41k	3%	98.1%

D. ABC class concentration

Class	SKUs	Inventory value share	Lost-sales share
Class A	24	33%	58%
Class B	36	32%	35%
Class C	60	34%	8%

E. Top SKU-location priorities

SKU-location	Opportunity (12m proxy)	Share of SKU pool
Health Product 16 @ WH-MAD	€158k	18%
Health Product 16 @ WH-LIS	€139k	16%
Health Product 16 @ WH-PORTO	€123k	14%
Health Product 16 @ WH-LYON	€110k	13%
Health Product 8 @ WH-LYON	€42k	5%
Health Product 8 @ WH-MAD	€40k	5%
Personal Care Product 3 @ WH-MAD	€36k	4%
Pet Care Product 31 @ WH-LYON	€28k	3%

F. Metric definitions

Term	Definition
Fill rate	Units fulfilled / units demanded.
Stockout rate	Units of unmet demand / units demanded.
Days of supply	Available inventory / average daily demand.
Lost-sales value	Unmet demand priced at unit selling price.
Lost-sales margin proxy	Lost-sales value at category gross-margin assumptions.
Excess inventory	Inventory value above the ABC days-of-supply policy cap.
Trapped working capital	Average daily inefficient inventory capital proxy.
Governance priority score	0-100 composite of service, stockout, excess, supplier, and working-capital risk.
12-month value pool	Recoverable lost-sales margin plus releasable working capital, proxy estimates.

G. Category-region segment risk (all 32 segments)

Composite governance priority score by category-region segment, with the main risk driver the score identifies for each. Ranked highest priority first.

Segment	Score	Tier	Main driver	Fill
Health | France South-East	52	Medium	Stockout Risk	91.3%
Health | Portugal North	48	Medium	Stockout Risk	94.4%
Health | Spain Central	48	Medium	Stockout Risk	91.7%
Health | Portugal South	46	Medium	Stockout Risk	95.3%
Pet Care | France South-East	42	Medium	Stockout Risk	91.3%
Frozen | Spain Central	41	Medium	Stockout Risk	91.7%
Frozen | France South-East	40	Medium	Stockout Risk	92.6%
Pet Care | Spain Central	40	Medium	Stockout Risk	92.4%
Household | France South-East	39	Medium	Stockout Risk	92.8%
Household | Spain Central	36	Medium	Stockout Risk	94.8%
Pet Care | Portugal North	35	Medium	Stockout Risk	95.3%
Pet Care | Portugal South	33	Low	Stockout Risk	97.1%
Household | Portugal South	33	Low	Stockout Risk	97.3%
Household | Portugal North	33	Low	Stockout Risk	96.8%
Frozen | Portugal North	33	Low	Stockout Risk	96.3%
Frozen | Portugal South	32	Low	Stockout Risk	97.3%
Personal Care | Spain Central	31	Low	Stockout Risk	95.7%
Personal Care | France South-East	31	Low	Stockout Risk	95.8%
Dairy | France South-East	29	Low	Stockout Risk	96.4%
Beverages | France South-East	28	Low	Stockout Risk	96.6%
Dairy | Spain Central	28	Low	Stockout Risk	96.6%
Personal Care | Portugal North	28	Low	Stockout Risk	97.6%
Beverages | Spain Central	27	Low	Stockout Risk	97.4%
Personal Care | Portugal South	26	Low	Stockout Risk	98.6%
Dairy | Portugal North	26	Low	Stockout Risk	97.5%
Snacks | France South-East	26	Low	Stockout Risk	96.1%
Snacks | Spain Central	26	Low	Stockout Risk	96.8%
Dairy | Portugal South	25	Low	Stockout Risk	98.3%
Beverages | Portugal South	24	Low	Stockout Risk	99.3%
Beverages | Portugal North	24	Low	Stockout Risk	98.9%
Snacks | Portugal North	23	Low	Stockout Risk	98.2%
Snacks | Portugal South	23	Low	Stockout Risk	99.1%

H. Monthly service KPIs (24 months)

Network fill rate, stockout rate, and lost-sales value by month over the full observation window. This is the series behind Figures 1 and 2.

Month	Fill rate	Stockout rate	Lost-sales value
2024-01	99.6%	0.4%	€125k
2024-02	97.6%	2.4%	€443k
2024-03	97.4%	2.3%	€608k
2024-04	96.0%	3.4%	€751k
2024-05	96.3%	3.4%	€782k
2024-06	95.6%	4.1%	€940k
2024-07	95.3%	4.3%	€1.1M
2024-08	95.8%	4.0%	€925k
2024-09	95.9%	3.6%	€781k
2024-10	96.3%	3.2%	€850k
2024-11	95.7%	4.5%	€940k
2024-12	92.4%	7.2%	€1.8M
2025-01	96.8%	3.0%	€665k
2025-02	98.0%	2.2%	€358k
2025-03	97.6%	2.4%	€544k
2025-04	96.0%	3.4%	€842k
2025-05	96.2%	3.4%	€814k
2025-06	96.6%	3.3%	€701k
2025-07	95.0%	4.7%	€1.1M
2025-08	96.0%	4.1%	€798k
2025-09	96.0%	4.0%	€789k
2025-10	96.1%	3.5%	€932k
2025-11	95.6%	4.2%	€866k
2025-12	92.0%	7.9%	€1.9M

I. Most overstocked SKUs (releasable-capital candidates)

Products carrying the most days of supply, the working-capital side of the opportunity. These are the lines to trim first when redeploying capital to fast-moving Class A items.

Product	Category	ABC	Days of supply	Avg inventory value
Health Product 80	Health	C	231	€108k
Pet Care Product 95	Pet Care	C	91	€68k
Household Product 108	Household	C	87	€46k
Dairy Product 118	Dairy	C	75	€10k

Product	Category	ABC	Days of supply	Avg inventory value
Personal Care Product 75	Personal Care	C	69	€10k
Pet Care Product 71	Pet Care	C	66	€10k
Household Product 100	Household	C	61	€33k
Beverages Product 113	Beverages	C	60	€47k
Frozen Product 69	Frozen	C	59	€15k
Health Product 72	Health	C	51	€45k